

Gravitational Coherence Surfaces as Information Boundaries of Source Attribution

Parity, Entropy, and Stress-Energy Bookkeeping in Nested Gravitational Scenes

Daniel Scieszinski
danski2017@gmail.com
Independent Research Program

May 2026

Abstract

A gravitational coherence surface, or GCS, is a closed parity surface where a declared gravitational contribution reaches equality with its derived gravitational context under a declared readability rule. It is not a force boundary, causal horizon, material membrane, dynamical stability limit, or modification of general relativity. Its significance is more precise: it draws a lawful modular boundary for gravitational bookkeeping inside a continuous field.

This paper develops a speculative information-theoretic interpretation of that boundary. We argue that a GCS can be understood as a modular attribution surface: the location where source-resolved gravitational identity becomes maximally contested by contextual explanation. In a minimal normalized support model, binary attribution entropy is maximized exactly at the parity condition $Q_A = Q_K$, where Q_A is the declared contribution and Q_K is its derived context. This gives a formal bridge between contribution-context parity and source-attribution uncertainty.

We then extend the interpretation to nested gravitational scenes. Since gravitational information is generated by stress-energy structure, a GCS may mark where resolved stress-energy detail changes bookkeeping scale: source-resolved inside, attribution-contested at parity, and context-assimilated outside. This does not alter the stress-energy tensor or the Einstein field equation. It concerns how a declared model determines when detailed source structure must be carried explicitly and when it may be represented effectively in a parent context.

Finally, we outline a dynamic extension in which GCS surfaces become worldsheets and causally bounded data pockets reconstruct those worldsheets differently under retarded access. The framework does not claim information destruction, new observables, physical membranes, or new gravitational dynamics. It proposes a formal language for asking where gravitational source identity remains attributable, where it becomes ambiguous, and where it may be lawfully compressed into context.

Keywords: general relativity, gravitational coherence surface, source attribution, parity boundary, information theory, entropy, stress-energy, nested scenes, coarse graining, gravitational readability

1 Introduction

Gravity does not stop at a boundary, but source identity may cease to be cleanly attributable across one.

This paper begins from that distinction. In general relativity, gravitation is described by metric structure, curvature, and stress-energy. The gravitational field is continuous in ordinary weak-field

settings, and no surface introduced here interrupts that continuity. Matter, light, and test particles do not bounce off a gravitational coherence surface. No new force is proposed.

Yet every practical gravitational model begins by declaring a scene. A scene has a source roster, coordinate convention, approximation regime, contribution of interest, parent context, and comparison rule. Once this declaration is made, a question appears that is not answered merely by saying that gravity is continuous:

Where does a declared gravitational contribution remain readable as itself against the gravitational context in which it is embedded?

The gravitational coherence surface answers that question by drawing a parity boundary. Its surface nature is not incidental. The surface is the act. It is the lawful line through which a continuous gravitational field becomes modularly bookkeepable.

The purpose of this paper is to formalize the information-theoretic meaning of that line. The proposal is simple: a GCS marks the boundary where source-resolved gravitational information and context-resolved gravitational information reach attribution parity. Inside the boundary, the source contribution dominates the bookkeeping. At the boundary, source and context hold equal claim. Outside the boundary, the source is increasingly assimilated into parent context.

The field remains continuous. The attribution changes.

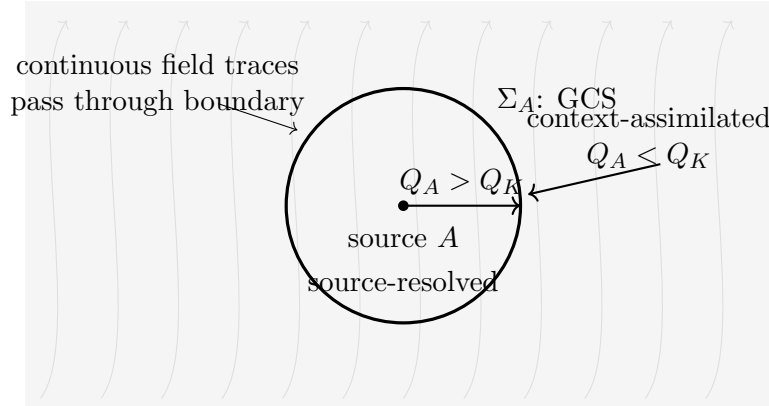


Figure 1: The original boundary act. A GCS does not stop the field. It marks the closed parity surface where a declared contribution and its derived context have equal bookkeeping claim.

2 GCS definition and boundary primacy

Let a declared gravitational scene contain a contribution A and a derived context K . A readability rule assigns comparable quantities to each:

$$Q_A(x) \tag{1}$$

and

$$Q_K(x). \tag{2}$$

The gravitational coherence surface associated with A is

$$\Sigma_A = \{x : Q_A(x) = Q_K(x)\}. \tag{3}$$

Equivalently, define the residual

$$\Psi_A(x) = Q_A(x) - Q_K(x), \quad (4)$$

so that

$$\Sigma_A = \{x : \Psi_A(x) = 0\}. \quad (5)$$

The meaning of this surface depends on the declared scene. A GCS is not a property of a source alone. It is a property of a contribution-in-context under a declared rule. Change the parent context, source roster, contribution type, or readability rule, and the surface may move or change shape.

This dependence is not a weakness. It is the core of the object. A source does not possess a single universal gravitational identity boundary in isolation from the scene in which it is read. It has contribution-context boundaries. These boundaries record where the source remains attributable as itself and where it becomes contextually absorbed.

Definition 1 (Boundary primacy). *A GCS is first a closed parity surface. Its ability to draw a lawful modular boundary is the origin of its significance. Higher structures such as parity networks, junctions, worldsheets, and reconstruction families inherit their meaning from this surface act.*

The GCS is not important because it is a new physical entity. It is important because it allows a model to ask where source identity can be kept distinct without pretending that gravity itself stops.

3 A GCS as a modular attribution boundary

Consider the sign of the residual

$$\Psi_A(x) = Q_A(x) - Q_K(x). \quad (6)$$

Where

$$\Psi_A(x) > 0, \quad (7)$$

the declared contribution has stronger readability than its context. Where

$$\Psi_A(x) = 0, \quad (8)$$

the declared contribution and derived context are in parity. Where

$$\Psi_A(x) < 0, \quad (9)$$

the context has stronger readability than the declared contribution.

If the surface Σ_A closes around a region D_A , then D_A can be interpreted as the source-resolved bookkeeping domain of A , subject to validation. The GCS itself is the attribution boundary of that domain.

This is a bookkeeping statement, not a force statement. A closed GCS says that a declared contribution has a coherent modular domain relative to its context. It gives the model permission, under the declared rule, to distinguish “inside the contribution’s readable domain” from “outside, where context has equal or greater claim.”

Closure is therefore not a visual preference. It is what makes modular attribution possible. If a computed surface is open, leaking, or perforated, the model has not automatically discovered an exotic object. Several failures must be ruled out first: incomplete scene declaration, insufficient domain, inadequate sampling, wrong context, projection artifact, tolerance artifact, or failed extraction. A genuinely multiply connected or topology-changing retained structure may be possible, but it must be reclassified and audited. It does not inherit the simple meaning of a closed GCS enclosure for free.

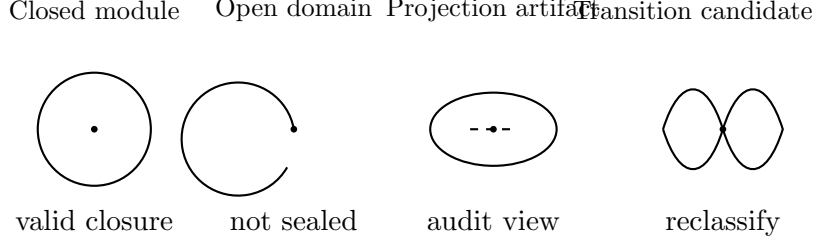


Figure 2: Closure and bookkeeping legitimacy. A simple closed GCS supports modular attribution. Open, apparently perforated, or topology-changing structures require audit before interpretation.

4 From contribution parity to attribution entropy

The GCS condition is a contribution-context equality. To connect it to information theory, we ask whether this equality also corresponds to maximum uncertainty in source attribution.

Let $Q_A(x)$ and $Q_K(x)$ be nonnegative supports for two attribution hypotheses:

- (i) the local gravitational information is best attributed to contribution A ;
- (ii) the local gravitational information is best attributed to context K .

Define normalized attribution weights:

$$w_A(x) = \frac{Q_A(x)}{Q_A(x) + Q_K(x)} \quad (10)$$

and

$$w_K(x) = \frac{Q_K(x)}{Q_A(x) + Q_K(x)}. \quad (11)$$

Then

$$w_A(x) + w_K(x) = 1. \quad (12)$$

These weights are not claimed to be universal probabilities. They are a minimal support model. In a more mature theory, the supports could be derived from metric perturbations, tidal tensors, likelihoods, instrument models, retarded source histories, or other declared evidence. The normalized model is useful because it exposes the first formal bridge.

Define binary attribution entropy:

$$S_{\text{attr}}(x) = -w_A(x) \log w_A(x) - w_K(x) \log w_K(x). \quad (13)$$

This entropy measures uncertainty over attribution between source and context. It is not thermodynamic entropy by default. It is uncertainty over bookkeeping explanation.

Proposition 1 (Binary attribution entropy is maximized at GCS parity). *Let $Q_A, Q_K > 0$, and define*

$$w_A = \frac{Q_A}{Q_A + Q_K}, \quad w_K = \frac{Q_K}{Q_A + Q_K}. \quad (14)$$

Let

$$S_{\text{attr}} = -w_A \log w_A - w_K \log w_K. \quad (15)$$

Then S_{attr} is maximized exactly when

$$Q_A = Q_K. \quad (16)$$

Proof. Binary Shannon entropy is maximized when

$$w_A = w_K = \frac{1}{2}. \quad (17)$$

Because

$$w_A = w_K \quad (18)$$

if and only if

$$Q_A = Q_K, \quad (19)$$

the entropy maximum occurs exactly at contribution-context parity. Therefore, in the normalized binary support model, the GCS coincides with maximal attribution uncertainty between the declared contribution and its derived context. $\square$

This result is deliberately narrow. It does not show that every physical GCS carries thermodynamic entropy. It does not make the surface an observable. It shows that the parity condition has a natural information-theoretic interpretation: the GCS is the boundary where source and context are equally supported as explanations.

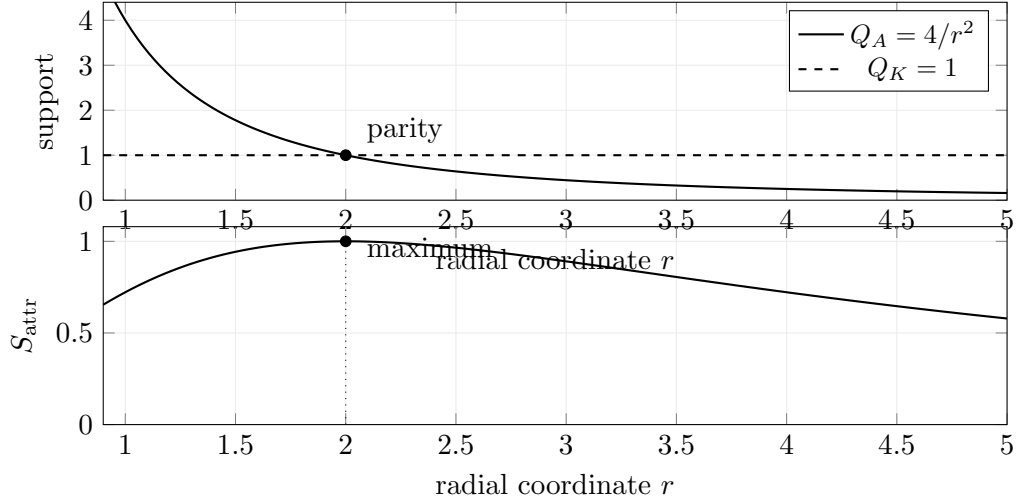


Figure 3: Entropy peak at parity. In the minimal normalized binary support model, attribution entropy is maximized exactly where the declared contribution and derived context reach equality.

5 Source-resolved, contested, context-assimilated

The normalized model gives a clean interpretation of the three GCS regions. Inside the GCS,

$$Q_A > Q_K \quad (20)$$

so

$$w_A > w_K. \quad (21)$$

The source attribution dominates. At the GCS,

$$Q_A = Q_K \quad (22)$$

so

$$w_A = w_K. \quad (23)$$

Attribution is maximally contested. Outside the GCS,

$$Q_A < Q_K \quad (24)$$

so

$$w_A < w_K. \quad (25)$$

The source is increasingly assimilated into context.

This gives the central information-theoretic interpretation: a GCS is a source-attribution boundary. It marks where gravitational information generated by a source ceases to be more attributable to that source than to the derived context of the scene.

Influence may continue. Attribution changes class.

This difference is crucial. A source can contribute to the gravitational field outside its GCS. The GCS does not terminate the field. It marks where the source's contribution no longer carries dominant source-resolved bookkeeping identity under the declared rule.

This is the core modulation mechanism. The GCS modulates attribution, not gravitational existence.

6 Many-source attribution and low-margin regions

The binary case is the cleanest. A many-source scene introduces additional structure. Let the scene contain readable supports

$$Q_i(x), \quad i = 1, \dots, N. \quad (26)$$

Define normalized attribution weights

$$w_i(x) = \frac{Q_i(x)}{\sum_{j=1}^N Q_j(x)}. \quad (27)$$

The many-source attribution entropy is

$$S_{\text{attr}}(x) = - \sum_{i=1}^N w_i(x) \log w_i(x). \quad (28)$$

Pairwise parity between sources i and j occurs where

$$Q_i = Q_j, \quad (29)$$

or equivalently,

$$w_i = w_j. \quad (30)$$

But in a many-source scene, not every pairwise equality surface is meaningful. A formal equality between two weak contributors may occur in a region dominated by a third contribution. A raw zero set may have little interpretive rank. This is why relevance filtering is necessary.

The many-source interpretation should therefore distinguish parity faces, low-margin regions, junctions, transition regions, and branch candidates. In this language, low-margin regions are high-attribution-entropy regions. Junctions are concentrated multi-source attribution ambiguities. Candidate parity nodes are not physical objects, but locations where bookkeeping identity becomes locally difficult to order.

This gives parity networks a natural information role: they record the relational registry of source attribution in a declared scene.

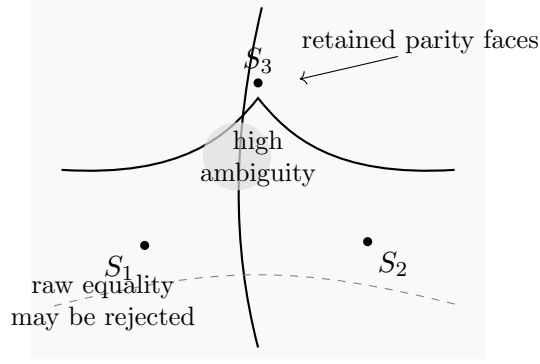


Figure 4: Many-source attribution. Retained parity faces and junction regions may be read as a relational registry of source attribution. Raw equalities need not be meaningful without relevance filtering.

7 Stress-energy and bookkeeping scale

The information interpretation becomes more significant when connected to stress-energy. In general relativity, source structure enters through the stress-energy tensor:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (31)$$

The GCS does not modify this equation. It does not change $T_{\mu\nu}$, add forces, or alter the metric. It is downstream of the declared source and field model.

However, if gravitational information is generated by stress-energy structure, then a boundary of source attribution is also a candidate boundary of stress-energy bookkeeping scale. Let a child scene C have resolved stress-energy structure

$$T_{\mu\nu}^C. \quad (32)$$

A parent context Π may represent that child through an effective or compressed description

$$\bar{T}_{\mu\nu}^{C|\Pi} = \mathcal{C}_\Pi[T_{\mu\nu}^C], \quad (33)$$

where $\mathcal{C}_\Pi$ is a declared compression or coarse-graining map.

Inside the child GCS, the resolved internal source structure of C remains relevant to the declared bookkeeping. At the GCS, the child contribution and parent/context claim reach parity. Outside the GCS, the child may be represented as an effective contribution to the parent context, if validation supports that compression.

Hypothesis 1 (Stress-energy bookkeeping scale). *A GCS may mark where resolved stress-energy detail becomes eligible for parent-context compression.*

This is not physical erasure. The detailed child scene remains present in a more complete description. But the parent model may not need to carry every internal detail explicitly once the child's contribution has become context-assimilated under the declared rule.

Thus the GCS may be interpreted as a compression boundary: inside, resolved stress-energy; at parity, attribution-contested stress-energy information; outside, effective stress-energy representation. This is a modeling statement, not a new law of nature.

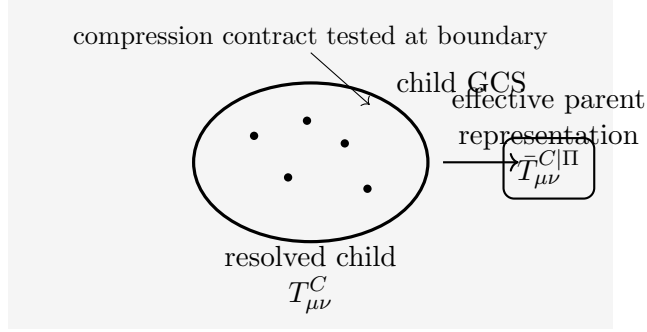


Figure 5: Stress-energy bookkeeping scale. A closed and validated GCS may mark where resolved child structure becomes eligible for effective parent-context representation.

8 Nested scenes and parent context

The stress-energy interpretation naturally leads to the nesting problem. A gravitational scene is rarely isolated. A moon is interpreted inside a planet system. A planet system is interpreted inside a stellar system. A stellar system is interpreted inside a galactic environment. Each level may obey the same local laws, while the bookkeeping changes across scale.

The GCS provides a candidate grammar for this handoff. It marks where a child contribution remains source-resolved and where it can be assimilated into a parent description. This does not mean the child vanishes. It means the parent may read the child differently.

A child scene can be internally high-resolution while its parent context is coarser, delayed, or incomplete. The handoff between child and parent is therefore not automatic. It requires a declared parent relation, a context rule, and validation.

In the present interpretation, the parent relation is a compression contract. It tells the model how child detail is inherited by the parent scene. The GCS is the surface where that contract is tested.

If the boundary closes and validates, the child can be bookkept as a coherent module. If the boundary fails, the child-parent relation is underdeclared or unstable.

This gives a sharper version of the nesting problem:

When is it lawful to replace resolved child stress-energy structure with parent-level effective context?

The answer cannot be “whenever the modeler wants.” The GCS offers a candidate diagnostic surface for when such replacement becomes legitimate.

9 Dynamic extension: GCS worldsheets

A static GCS is a surface on a declared slice:

$$\Sigma_A(t) = \{x : \Psi_A(x, t) = 0\}. \quad (34)$$

In a time-dependent scene, the corresponding object is a worldsheet:

$$\mathcal{W}_A = \{(x, t) : \Psi_A(x, t) = 0\}. \quad (35)$$

This worldsheet is not a material object. It is the history of a parity boundary.

If a GCS is an attribution boundary, then $\mathcal{W}_A$ is the history of an attribution transition. It records where and when source identity remains resolved, becomes contested, or is assimilated into context.

Dynamic scenes add causal delay. Gravitational information propagates causally. A baseline model may compute an ideal scene at a coordinate time, but a situated data pocket receives only what its causal access permits.

Let p be a causally bounded data pocket. Let

$$\mathcal{S}_p(\tau) \tag{36}$$

be the scene reconstructed by pocket p at local update parameter τ . The pocket extracts an apparent GCS

$$\Sigma_A^{(p)}(\tau) = \{x : \Psi_A^{(p)}(x, \tau) = 0\}. \tag{37}$$

The family of such reconstructed scenes forms a higher-order object:

$$\mathfrak{S}(\mathcal{S}) = \{\mathcal{S}_p(\tau)\}. \tag{38}$$

This is the scene-of-scenes.

The scene-of-scenes is not a metaphysical claim. It is the family of admissible reconstructions of a declared model under bounded causal access.

A comparison between a baseline GCS and a pocket-reconstructed GCS is meaningful only after identity tracking is earned. If P_0 is a baseline parity candidate and P_p is a pocket-reconstructed candidate, one may write

$$\Delta_p = P_p - P_0 \tag{39}$$

only when P_p has been shown to be the same candidate under the declared tracking criteria. Otherwise, a large offset may indicate branch ambiguity, solver artifact, multiple low-residual regions, or parent-context divergence.

No identity tracking, no meaningful reconstruction shear.

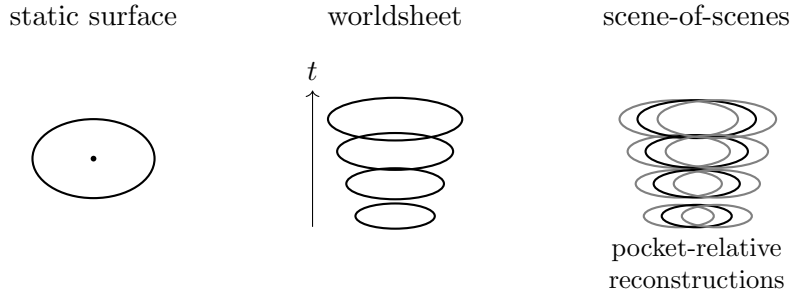


Figure 6: From surface to worldsheet to scene-of-scenes. A dynamic GCS is a parity boundary extended through time. Causally bounded data pockets may reconstruct shifted or branched versions of the same boundary, subject to identity tracking.

10 Agency as a downstream implication

Agency is not required for the core argument, but the framework naturally points toward it. A passive data pocket reconstructs a scene. An agentic data pocket reconstructs a scene and acts.

The agent does not act from omniscient access. It acts from its own reconstruction, with delayed information, compressed memory, attribution uncertainty, and incomplete parent context.

Let an agentic pocket p reconstruct

$$\mathcal{S}_p(\tau) \tag{40}$$

and select an action

$$a_p \in \mathcal{A}_p(\tau). \tag{41}$$

The action updates future source history through ordinary physical channels:

$$\mathcal{S}_p(\tau) \rightarrow a_p \rightarrow \mathcal{S}(t + \Delta t). \tag{42}$$

No new physics is implied. The internal model does not bend spacetime by intention. It selects action through a physical body or mechanism, and that action becomes part of the future scene.

In this language, agency is action under bounded reconstruction. This belongs to a later paper. It is included here only to show that once attribution entropy and data pockets are admitted, agency becomes a natural extension. Agents act before uncertainty is gone. They act from reconstructed context, and their actions alter future histories.

11 Claim ladder

The framework must be separated into levels.

Level	Claim status
1	A GCS is a declared contribution-context parity surface.
2	A closed GCS functions as a modular attribution boundary.
3	Under normalized binary support, attribution entropy is maximized exactly at GCS parity.
4	A GCS may mark where source-specific gravitational information becomes attribution-contested against context.
5	A GCS may mark where resolved stress-energy detail changes bookkeeping scale relative to parent context.
6	A GCS worksheet and its data-pocket reconstructions form a scene-of-scenes structure.
7	Agentic data pockets act from reconstructed scenes under attribution uncertainty, altering future source history through ordinary physical channels.

Table 1: Claim ladder. Only Levels 1 through 3 are formalized in this paper. Levels 4 through 7 are a research program.

Only Levels 1 through 3 are formalized in this paper. Levels 4 through 7 are a research program.

12 Non-claims

This paper does not modify general relativity. It does not introduce a new force. It does not introduce a new field equation. It does not claim that GCS surfaces are physical membranes, causal horizons, or dynamical stability boundaries. It does not claim that matter, light, or test particles

are blocked by a GCS. It does not claim that information is destroyed at the GCS. It does not claim that attribution entropy is thermodynamic entropy. It does not claim that parity nodes are observables. It does not claim that reconstruction shear is automatically physical. It does not claim that stress-energy changes at the GCS. It does not claim that coarse-graining is automatically valid outside any apparent boundary.

The claim is narrower: a GCS may be interpreted as a lawful modular boundary of source attribution, and in a minimal normalized binary support model, that boundary coincides with maximal attribution entropy.

Everything beyond that requires testing.

13 Toy-model research program

The proposed interpretation should be easy to weaken. If entropy adds nothing, the branch should be retired. If it predicts stable diagnostics, it may become useful.

13.1 Binary source-context entropy test

Declare a source contribution Q_A and derived context Q_K . Compute

$$w_A = \frac{Q_A}{Q_A + Q_K}, \quad w_K = \frac{Q_K}{Q_A + Q_K}, \quad (43)$$

and

$$S_{\text{attr}} = -w_A \log w_A - w_K \log w_K. \quad (44)$$

Verify that the entropy ridge coincides with the GCS surface $Q_A = Q_K$.

13.2 Many-source entropy test

For three or more sources, compute

$$w_i = \frac{Q_i}{\sum_j Q_j} \quad (45)$$

and

$$S_{\text{attr}} = -\sum_i w_i \log w_i. \quad (46)$$

Compare entropy ridges with retained parity faces, low-margin regions, and junction candidates. The many-source entropy field should not reduce to any one pairwise surface. It should highlight multi-source attribution ambiguity.

13.3 Child-parent compression test

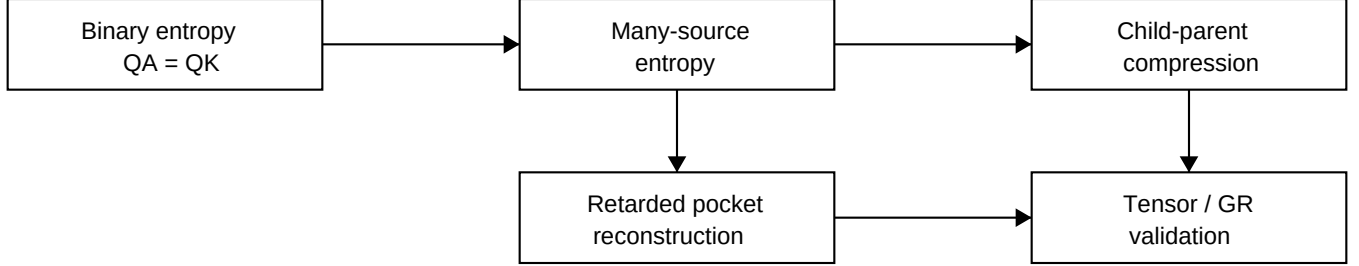
Declare a child scene C and parent context Π . Compare resolved child support against parent/context support. Then compare a resolved child model against an effective compressed model. The question is where compression preserves relevant diagnostics and where it fails.

13.4 Retarded data-pocket reconstruction test

Let a source move or change state. Compare a baseline GCS to a causally delayed pocket-reconstructed GCS. The question is whether parity candidates remain local continuations, jump to other candidate basins, or become identity-ambiguous.

13.5 Tensor validation test

Replace or supplement scalar support with GR-native diagnostics: tidal tensor, electric-Weyl structure, metric perturbation, geodesic deviation, or curvature summaries. The question is whether entropy and parity structures correlate with field quantities beyond scalar scaffolds.



Each step may fail. Failure is evidence, not embarrassment.

Figure 7: Toy-model test ladder. The entropy interpretation should be hardened through small tests before any stronger interpretation is promoted.

14 Discussion

The GCS began as a surface. This paper argues that its surface nature is exactly why it matters. A continuous gravitational field can still contain lawful modular boundaries of interpretation. These boundaries need not be physical membranes. They need not block motion. They need not introduce new dynamics. They can still be mathematically powerful because they determine when a declared contribution remains readable as itself.

The information-theoretic interpretation sharpens this point. If readability supports are interpreted as attribution supports, parity is exactly the point of maximal binary attribution uncertainty. This gives the GCS a natural role as an information boundary of source attribution.

The stress-energy extension is speculative but structurally motivated. Since gravitational information originates in stress-energy structure, a boundary of source attribution may also mark where resolved stress-energy detail becomes eligible for parent-context compression. This does not alter the stress-energy tensor. It governs how much of that tensor's structure must remain explicit in a declared model.

The dynamic extension adds causal delay. A static GCS becomes a worldsheet. Bounded data pockets reconstruct that worldsheet differently. The family of reconstructions becomes a scene-of-scenes. Stability across that family may help determine whether a parity boundary is only an ideal baseline object or a robust feature of causal reconstruction.

The same original boundary act supports all three ideas: attribution, compression, and reconstruction.

15 Conclusion

A gravitational coherence surface is the lawful line-drawing operator of gravitational readability. It is a surface, and that is its power.

It does not stop gravity. It does not alter general relativity. It does not introduce a new field, force, or physical membrane. It marks where a declared gravitational contribution reaches parity with its derived context, allowing a continuous field to support modular bookkeeping.

This paper proposed that the same surface has a natural information-theoretic interpretation. In a minimal normalized support model, the GCS condition

$$Q_A = Q_K \tag{47}$$

is exactly the condition of maximal binary attribution entropy. The surface therefore marks where source identity is most contested by context.

From there, a cautious staircase appears:

GCS parity gives modular attribution. Modular attribution gives source-information boundaries. Source-information boundaries touch entropy. Entropy touches compression. Compression touches effective stress-energy bookkeeping. Dynamic scenes turn surfaces into worldsheets. Causal data pockets turn worldsheets into reconstruction families. Agents, if included, act from bounded reconstructions and alter future source histories through ordinary physical channels.

The field remains continuous. The bookkeeping changes.

The GCS is where gravitational identity becomes an accounting problem.

References

- [1] T. Buchert, “On Average Properties of Inhomogeneous Cosmologies,” *General Relativity and Gravitation*, vol. 32, pp. 105-125, 2000.
- [2] G. F. R. Ellis, “Relativistic Cosmology: Its Nature, Aims and Problems,” in *General Relativity and Gravitation*, 1984.
- [3] A. Einstein, “Die Feldgleichungen der Gravitation,” *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, 1915.
- [4] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation*. W. H. Freeman, 1973.
- [5] R. Penrose, *Techniques of Differential Topology in Relativity*. SIAM, 1972.
- [6] D. Scieszinski, “The Gravitational Coherence Surface: Parity Horizons of Gravitational Identity,” Independent Research Program, 2026.
- [7] D. Scieszinski, “Many-Source Parity Networks in Weak-Field Gravitational Readability,” Independent Research Program, 2026.
- [8] D. Scieszinski, “Mesoscale Gravitational Readability: Derived Interface Geometry Between Local Evaluation and Coarse Averaging,” Independent Research Program, 2026.
- [9] D. Scieszinski, “Proof-of-Concept Extraction of Many-Source Gravitational Readability Structure,” Independent Research Program, 2026.
- [10] D. Scieszinski, “The Nesting Domain Handoff Problem: Parity Networks and the Modular Structure of Gravitational Scenes,” Independent Research Program, 2026.
- [11] C. E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, vol. 27, pp. 379-423 and 623-656, 1948.
- [12] R. M. Wald, *General Relativity*. University of Chicago Press, 1984.